

Enzyme specificity prediction using cross-attention graph neural networks

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Enzymes are the molecular machines of life, and a key property that governs their function is substrate specificity—the ability of an enzyme to recognize and selectively act on particular substrates. This specificity originates from the three-dimensional (3D) structure of the enzyme active site and complicated transition state of the reaction^{1,2}. Many enzymes can promiscuously catalyse reactions or act on substrates beyond those for which they were originally evolved^{1,3–5}. However, millions of known enzymes still lack reliable substrate specificity information, impeding their practical applications and comprehensive understanding of the biocatalytic diversity in nature. Here we developed a cross-attention-empowered SE(3)-equivariant graph neural network architecture named EZSpecificity for predicting enzyme substrate specificity, which was trained on a comprehensive, tailor-made database of enzyme–substrate interactions at sequence and structural levels. EZSpecificity outperformed the existing machine learning models for enzyme substrate specificity prediction, as demonstrated by both an unknown substrate and enzyme database and seven proof-of-concept protein families. Experimental validation with eight halogenases and 78 substrates showed that EZSpecificity achieved a 91.7% accuracy in identifying the single potential reactive substrate, significantly higher than that of the state-of-the-art model enzyme substrate prediction (58.3%). EZSpecificity represents a general machine learning model for the accurate prediction of substrate specificity for enzymes related to fundamental and applied research in biology and medicine.

Over the past few years, several machine learning tools have been developed to predict enzyme substrate specificity, but they met with limited success and mainly concentrated on developing computational tools specific to particular protein families^{6–12}. Popular enzyme function prediction tools such as contrastive learning-enabled enzyme annotation (CLEAN)¹³, ProteInfer¹⁴ and DeepECTransformer¹⁵ struggle to distinguish enzyme reactivity and substrate specificity within the same Enzyme Commission (EC) numbers, which is a central challenge for biocatalysis. Enzyme substrate prediction (ESP) used graph neural networks (GNNs) to encode various metabolites for numerous proteins⁸. However, ESP faced challenges in covering all native and non-native enzyme–substrate pairs in a genome-scale metabolic model because of the limited number of collected substrates (about 1.3k). Fingerprint and sequence-based embeddings have since achieved more promising results^{8,13,16,17}, although compound–protein interaction (CPI) methods still struggle to accurately capture substrate–enzyme recognition⁷. This reflects the importance of catalytic residue identity, spatial arrangement and long-range coupling in determining specificity^{18–20}. Although the frameworks such as ESP, CPI and ALDELE (all-purpose deep-learning-based multiple toolkit)²¹ highlight the potential for

broader applicability, an important challenge remains as to how to fully integrate and leverage information from sequences, structures and interactions simultaneously to enhance predictive performance.

To address the above-mentioned limitations, we developed a general deep learning model named EZSpecificity for enzyme specificity prediction, which is empowered by structural information and a cross-attention-assisted GNN. We first built a high-quality curated enzyme–substrate interaction dataset (ESIBank) encompassing a wide range of natural and non-natural substrates, substitutions and substrate–enzyme complex information at the sequence, structure and interaction levels. We then designed an SE(3)-equivariant GNN-based encoder to capture the micro-environment of each atom in the catalytic active site, followed by a cross-attention process aimed at empowering interactions between the substrates and the enzymes. To validate the versatility and accuracy of EZSpecificity, we extensively performed *in silico* experiments for various unknown enzyme and substrate combinations, as shown for six typical protein families. Last, we challenged EZSpecificity to annotate substrate landscape for an in-house collected halogenase database (386 enzymes and 449 substrates), followed by experimental validation.

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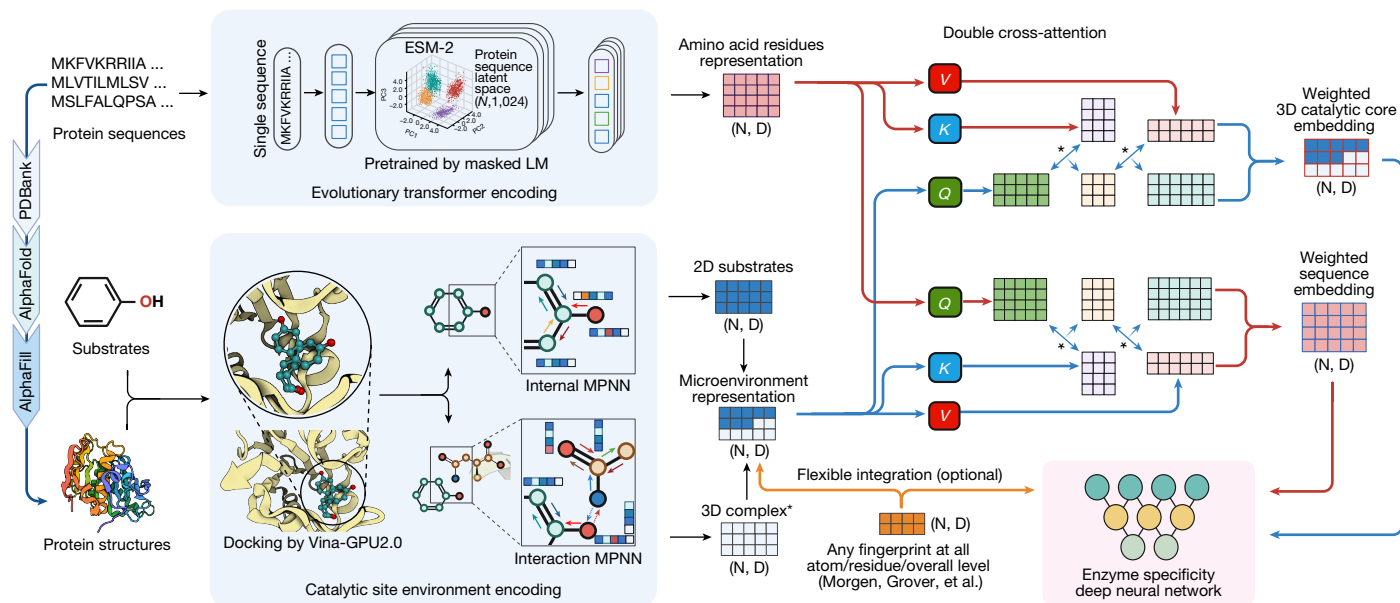


Fig. 1 | The machine learning architecture of EZSpecificity. ESM-2 and active site environment encoding method empowered by SE(3)-equivariant GNN were applied to extract the sequence and structural embedding of the enzyme–substrate complex. Message passing neural networks (MPNN) were used for obtaining the embedding of the internal substrate and the embedding of enzyme–substrate interaction. The cross-attention layer contained the independent heads, which were a learnable attention function to capture the

interaction between the amino acids of enzymes and the atoms of substrates. The attention function was composed of several trainable linear layers. Weighted 3D-catalytic core embedding and weighted sequence embedding were used as input for training the multi-layer perceptron model. Any fingerprint at the atom, residue and overall levels can be flexibly integrated into the embedding process. Schematic was created using BioRender (<https://biorender.com>).

The architecture of EZSpecificity

Existing machine learning models for enzyme specificity prediction mainly concentrated on extracting features from one-dimensional protein sequences and two-dimensional molecular graphs, ignoring the three-dimensional (3D) nature of the substrate binding process (Supplementary Table 1). Moreover, these models reduced the substrate and enzyme to two separate embeddings before the concatenation steps, which probably limited their ability to capture complex interactions between the enzyme and substrate²². To address these challenges, we developed a deep learning model named EZSpecificity that integrates sequence data, 3D enzyme–substrate complex structures and active site environment to predict enzyme–substrate specificity. The architecture of EZSpecificity is shown in Fig. 1. The full-length amino acid representations produced by ESM-2, one of the most powerful self-supervised sequence representation encoders, were integrated into EZSpecificity at the sequence level. Considering that the substrate binding pocket is crucial for determining enzyme function, especially specificity, we represented the binding pocket as a graph $G = (V, E)$, where each node V is an atom from either substrate or enzyme. These nodes are connected by edges in E to their k -nearest atom neighbour in the structure. To maintain the invariant of the encoding process to E(3) transformations, drawing inspiration from E(n)-equivariant graph neural networks²³, the binding pocket is modelled using an SE(3)-equivariant GNN. Furthermore, we kept the gate open for flexibly integrating other fingerprints at the atom, residue or overall level into the framework. The above design allowed the model to incorporate essential features, thereby enhancing the generalizability of EZSpecificity. Importantly, EZSpecificity is designed to predict enzyme substrate specificity, but, at present, it does not support reliable prediction of chemo-, regio- or stereoselectivity because of the limitations in molecular representation and the resolution of the encoding strategy.

To capture the complex enzyme–substrate interaction in the active site environment, two cross-attention layers were used to generate

the blended representation of the enzyme–substrate pair. The cross-attention layers can emphasize the amino acids and atoms crucial for enzyme–substrate specificity, unlike previous approaches that assigned equal weight to each. Our approach can minimize unnecessary noise and enhance the focus of the model, thereby improving its generalizability. Overall, the cross-attention layer empowered the interactions between the amino acids of the enzyme and the atoms of the substrate, potentially capturing significant specificity-driving information. Finally, a multi-layer perceptron is used to predict the specificity based on the representation of the enzyme–substrate pair. Subsequently, EZSpecificity was trained using the ESIBank dataset, as described in the next section. The main input data included enzyme sequences, enzyme–substrate complexes from docking and substrate SMILES. The model generates a prediction score that defines the enzyme specificity.

Enzyme–substrate interaction database

A high-quality training dataset plays an important part in enhancing the performance of machine learning models. To train the EZSpecificity model, we sought to develop a comprehensive enzyme–substrate interaction database (ESIBank) by integrating the sequence-based databases (for example, BRENDA and UniProt) with structural information generated by AlphaFold, AlphaFill and AutoDock/Vina-GPU (Fig. 2a and Supplementary Figs. 1 and 2). After obtaining diverse enzyme sequences from BRENDA and UniProt, we also collected enzyme–substrate data for six representative enzyme families, including esterases²⁴, glycotransferases²⁵, nitrilases²⁶, phosphatases²⁷, thiolases⁶ and domain of unknown function proteins (DUFs)²⁸. Knowledge bases such as UniProt use expert curation progress and rule-based prediction systems to gather data from various sources²⁹, whereas their update often lags years behind the rate at which new experimental data are released. To capture crucial enzymatic reactions shown in images but not yet indexed by public databases, we developed a semi-automatic data extraction process consisting of four steps:

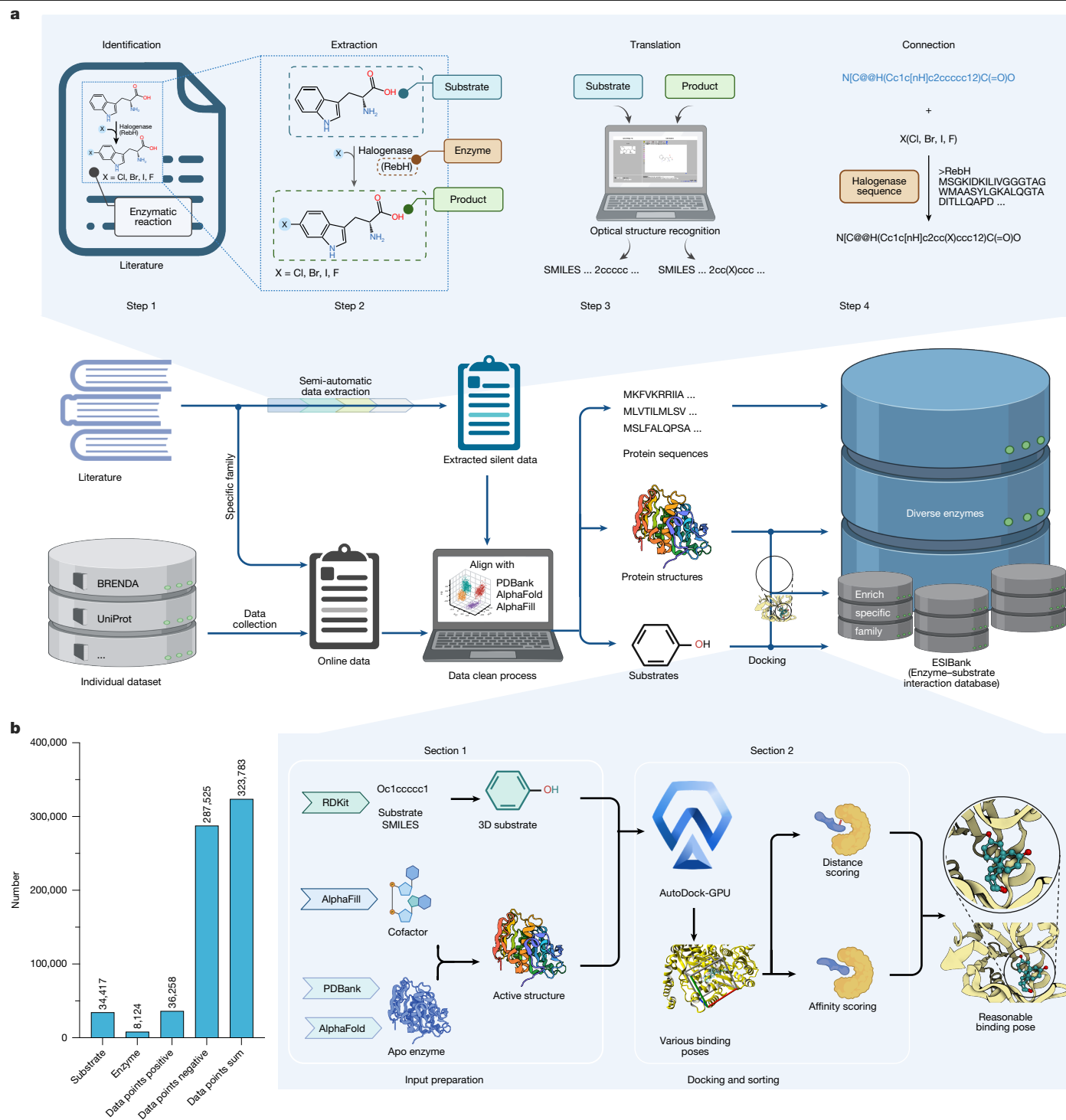


Fig. 2 | Construction of the comprehensive enzyme–substrate interaction (ESIBank) database. **a**, The process of constructing the ESIBank database. The middle panel described the whole process of obtaining various natural and non-natural substrates and wild-type and variant enzymes from published online servers and literature. The semi-automatic data extraction and AutoDock-GPU docking process were presented in the top and bottom panels, respectively. Semi-automatic data extraction includes four steps: identification, extraction, translation and connection. OSRA (Optical Structure Recognition Application) converts the chemical structure information of chemical compounds from

identification, extraction, translation and connection (Fig. 2a, top). As proof of concept, a halogenase–substrate (HaloS) dataset containing around 3,300 enzyme–substrate pairs was established using this process for later model validation (Supplementary Text 1).

scientific publications into machine-readable SMILES format. The halogenation reaction is chosen as the example for applying semi-automatic data extraction. AutoDock-GPU docking process includes two sections: (1) input preparation and (2) docking and sorting process. PDBank, AlphaFold and AlphaFill are the sources for obtaining the 3D structures of proteins. Various binding poses obtained from AutoDock-GPU were filtered by a distance and affinity scoring system to select the most catalytic enzyme–substrate pose. **b**, Overview of the collected ESIBank dataset. Schematic in **a**, **b** was created using BioRender (<https://biorender.com>).

Next, to prepare the functional 3D structure for docking, we used cofactors sourced from BRENDA as indicators to identify cofactor-involved structures from the AlphaFill database. Moreover, for all reactions with annotated catalytic residues in BRENDA, these catalytic

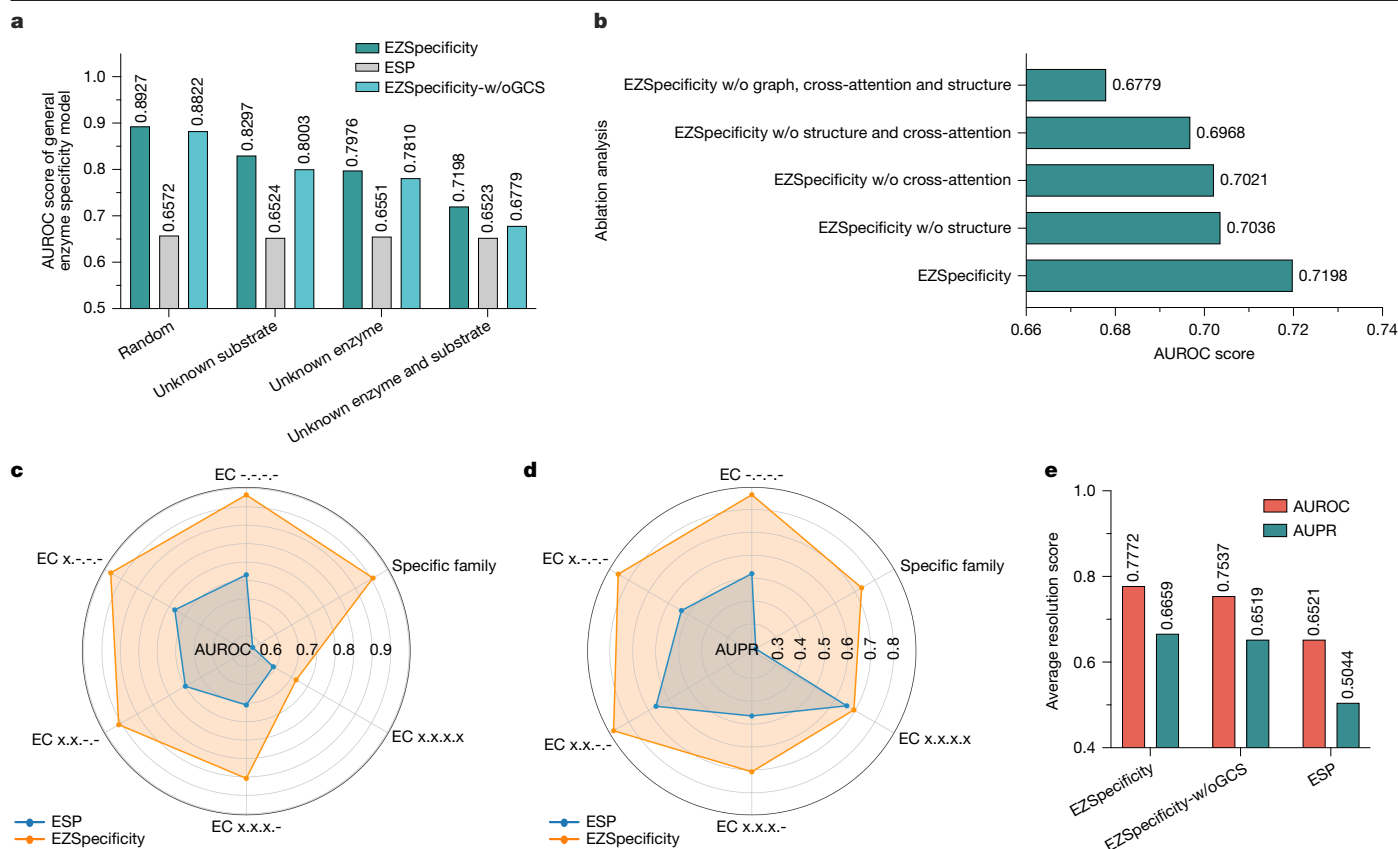


Fig. 3 | Evaluation of EZSpecificity on ESIBank dataset. a, The AUROC performance of EZSpecificity and the general enzyme specificity model ESP, evaluating on four dataset splits (random, unknown substrate, unknown enzyme and unknown enzyme and substrate). **b**, Ablation experiments of EZSpecificity on the unknown enzyme and substrate dataset by evaluating the AUROC score. **c, d**, The prediction resolution of EZSpecificity towards four digits of EC number by evaluating AUROC (**c**) and AUPR (**d**). The resolution was defined as the AUROC or AUPR score of models in the scenarios that were distinguished based on the number of shared EC number digits between the actual enzyme–substrate pair and its corresponding negative samples.

residues were used to define the active site for the docking box, and the substrate structure was prepared from Simplified Molecular Input Line Entry System (SMILES) using RDKit, followed by the docking process using AutoDock-GPU (Fig. 2a, bottom). AutoDock-GPU, as a hardware-accelerated version of AutoDock, enabled exploring the molecular space as necessary, while safely avoiding redundant computations^{30,31}. We applied distance scoring for those pairs with known active sites and affinity scoring to determine the most favourable binding pose over 100 runs, with each run allowing up to 2,500,000 score evaluations—totalling a potential maximum of 250 million pose evaluations per pair. In summary, ESIBank covered a wide range of natural and non-natural substrates (34,417), wild-type and variant enzymes (8,124) from published online servers and literature. The ESIBank database contains 25 times more substrates than the dataset available in ESP, which consists of 1,379 unique substrates⁸. In total, 323,783 enzyme–substrate pairs were compiled to generate ESIBank (Fig. 2b and Supplementary Texts 2 and 3), representing the first broad enzyme–substrate complex database at the structural level to date.

In silico evaluation of EZSpecificity

To validate EZSpecificity, we designed multiple benchmarking experiments to evaluate its overall effectiveness in predicting enzyme

Owing to the absence of EC number annotations in data from specific enzyme families, they are grouped into one category. **e**, Average resolution score on all four dataset splits. EZSpecificity-w/oGCS and ESP were also investigated for comparison. EZSpecificity without graph, cross-attention and structural embeddings is abbreviated as EZSpecificity-w/oGCS. AUPR and AUROC are more suited when the focus is on identifying positive instances in an imbalanced dataset, whereas MCC provides a balanced view across all classes. For tasks such as enzyme specificity prediction, in which the identification of positive hits is often a primary goal, AUPR and AUROC can be more informative and relevant.

specificity on full ESIBank (Fig. 3). EZSpecificity was first compared with ESP, the state-of-the-art general machine learning model for predicting enzyme substrate specificity. Four-fold cross-validation was performed on the ESIBank dataset, calculating the mean performance across all folds. Apart from the conventional setting of random splits, we assessed our model under three more conditions: (1) unknown substrate; (2) unknown enzyme; and (3) unknown enzyme and substrate. For example, the unknown substrate scenario includes exclusive substrates not found in the other folds. These conditions were designed to mimic various real-world scenarios. Notably, EZSpecificity consistently outperformed ESP across all four scenarios (Fig. 3a and Supplementary Fig. 3). As the difficulty increased with the use of unknown substrates and/or enzymes, the improvement of EZSpecificity or EZSpecificity-w/oGCS over ESP appeared to slightly diminish. However, it showed particularly impressive performance in the most challenging scenario, unknown enzyme and substrate (AUROC (EZSpecificity) = 0.7198 and AUROC (ESP) = 0.6523, where AUROC is the area under the receiver operating characteristic curve), indicating the improved reliability for practical applications. Moreover, an 8.72% improvement in the area under the precision versus recall curve (AUPR) score was observed (Supplementary Fig. 4).

It is important to acknowledge that, due to the limited accessibility of the ESP, the final ESP model was used directly for validation in

this study. EZSpecificity-w/oGCS and ESP share almost the same architecture but were trained on different datasets: ESIBank (about 323k data points, 34k substrates and 8k enzymes) for EZSpecificity-w/oGCS, and ESP-database (about 18k data points, 1.3k substrates and 12k enzymes) for ESP. As shown in Fig. 3a, the AUROC value of EZSpecificity-w/oGCS (0.8822) on the random dataset was significantly higher than that of ESP (0.6572). A similar trend was observed for the AUPR score, with EZSpecificity-w/oGCS achieving 0.5762 compared with 0.2057 for ESP. These results strongly indicate that the comprehensive and high-quality training dataset of ESIBank contributed significantly to the observed improvement in model performance. Moreover, the enhanced architecture, integrating both structural and interaction data, along with the expanded and comprehensive ESIBank, has a lot of potential for boosting performance on previous and/or newly released datasets.

Moreover, ablation analysis was investigated to dissect the performance of each component in EZSpecificity. As shown in Fig. 3b, the ablation experiment showed loss of active structure decreased the AUROC score from 0.7198 to 0.7036, demonstrating the benefits of explicitly modelling atomic interactions between the binding pocket of the enzymes and substrates. Similarly, loss of cross-attention layers decreased the AUROC score from 0.7198 to 0.7021, indicating that cross-attention layers have important roles in improving the model performance. Although these numerical changes may seem small, in the context of computational modelling, particularly in deep learning, even marginal improvements are challenging to achieve and can reflect meaningful enhancements in predictive accuracy. Beyond the ablation experiment, we observed that structural features, along with graph and cross-attention mechanisms, played a crucial part in enhancing the performance of EZSpecificity, especially on the halogenase dataset, a high-quality curated dataset that is described in the following section. In contrast to previous methods that used average pooling across all amino acids^{1,18,32}, the cross-attention layers enhanced the ability of the model to focus on the atoms and/or amino acids that determined enzyme substrate specificity, thereby reducing the bias introduced by indirect atoms and/or amino acids. The ablation analysis, which evaluated the AUPR score, showed similar results (Supplementary Fig. 4).

CPI was crucial to the discovery of new drugs by screening candidate compounds³³, holding the potential for predicting enzyme substrate specificity. After careful architecture comparison, we found the architecture of CPI is equivalent to EZSpecificity without graph, cross-attention and structural embeddings (EZSpecificity-w/oGCS). To achieve a fair comparison, CPI (also called EZSpecificity-w/oGCS) was retrained on the ESIBank and EZSpecificity, with full architecture outperforming EZSpecificity-w/oGCS in all examinations (Fig. 3a). Moreover, EZSpecificity demonstrated superior performance across all levels of resolution of EC number (Fig. 3c–e, Supplementary Table 3 and Supplementary Figs. 5–12), confirming the robustness and resolution of EZSpecificity at different depths in predicting enzyme substrate specificity. Furthermore, EZSpecificity slightly outperformed ESP under EC.x.x.x.x, highlighting its ability to distinguish among homologous enzymes and/or enzyme variants. The fine-tuning process with specific protein families could further enhance performance, as demonstrated later in Fig. 4d. The degree of improvement by EZSpecificity over ESP varied across different levels of the EC number, which may be attributed to the imbalance in dataset representation at varying depths of the EC hierarchy.

Generalizability across enzyme families

As models scale, efficiently and effectively adapting large models for specific downstream tasks emerges as a compelling research challenge³⁴. To explore the transferability of EZSpecificity to enzyme families of particular interest to researchers, we examined its performance on six enzyme families mentioned above (Fig. 4a). EZSpecificity performed well on all these enzyme families for three types of

datasets (random, unknown substrate and unknown enzyme) with up to 0.6835 averaged AUPR value (Fig. 4b and Supplementary Table 4). Moreover, we explored two fine-tuning strategies to determine the optimal approach for applying EZSpecificity to targeted enzyme families. The first action was to update the weights of the EZSpecificity model for several iterations, using data from the target enzyme family, which we referred to as the EZSpecificity-fine-tune model. As expected, the EZSpecificity model was further improved after fine-tuning actions and AUPR (EZSpecificity) on unknown substrate split increased about 7% compared with without fine-tuning (Fig. 4b and Supplementary Table 5). These observations agree well with the previous studies^{35–38}, in which the fine-tuning process ensured that the model understands and uses focused knowledge appropriately through reinforcing vocabulary. Similar results were observed for the AUROC evaluation (Supplementary Fig. 13).

The second action was to train a new specific model from scratch with the same architecture as EZSpecificity, using only data from the target enzyme family (positive samples ranging from 85 to 5,424), which was referred to as the EZSpecificity-individual model. The EZSpecificity-individual model surpassed the CPI-individual model in AUPR examination across all three dataset splits, showing an enhancement of 4.2–8.3% (Fig. 4c and Supplementary Table 6). The 27.4–54.5% improvement can be achieved when compared with ESP. These results indicate that EZSpecificity can effectively manage a limited number of data points, probably benefiting from the integration of 3D structural context, which simplifies the task of predicting enzyme specificity. Similar observations were also obtained for AUROC values within both fine-tuning strategies (Supplementary Fig. 14 and Supplementary Tables 5 and 6).

Furthermore, we challenged EZSpecificity on understudied enzymes and substrates (Fig. 4d and Supplementary Fig. 15). As shown in Fig. 4d, EZSpecificity surpassed CPI and ESP across all selected enzyme families. Fine-tuning does not invariably lead to improvements. Specifically, for thiolases and glycotransferases, its performance slightly decreased after fine-tuning (for example, AUPR (EZSpecificity, Thiolases) = 0.8247, AUPR (EZSpecificity-fine-tune, Thiolases) = 0.8230). Furthermore, the effectiveness of different fine-tuning actions varies, as shown by the performance on two enzyme families, DUFs and thiolases, in which starting from EZSpecificity was better than starting from scratch. A smaller model (for example, EZSpecificity-individual) trained on a limited dataset might outperform a larger model (EZSpecificity) because it is less prone to overfitting and can generalize better to new data, particularly when the task is straightforward or the data are of high quality. Conversely, for other enzymes, EZSpecificity-individual showed better results than both EZSpecificity and EZSpecificity-fine-tune. These findings demonstrated that the effectiveness of strategies for using EZSpecificity in downstream tasks differs in practical applications. Although it may be crucial to perform initial tests on individual evaluation subsets to identify the most effective approach for using EZSpecificity, directly applying EZSpecificity remains the recommended option when computer resources are limited.

Experimental validation of halogenases

Next, we sought to validate the prediction accuracy of EZSpecificity using halogenases, an understudied enzyme family with limited information on their substrate specificities. Halogenases were chosen for proof-of-concept because of their increased use in biocatalytic C–H functionalization and their excellent catalyst-controlled specificity and selectivity³⁹. Generally, small molecules with halogen atom(s) produced by halogenases have promising bioactivity and physicochemical properties, thereby offering broad application in pharmaceutical and agrochemical fields⁴⁰. By applying the semi-automatic data extraction method (Supplementary Text 1), 386 sequences were identified covering all four types of halogenases (haloperoxidases,

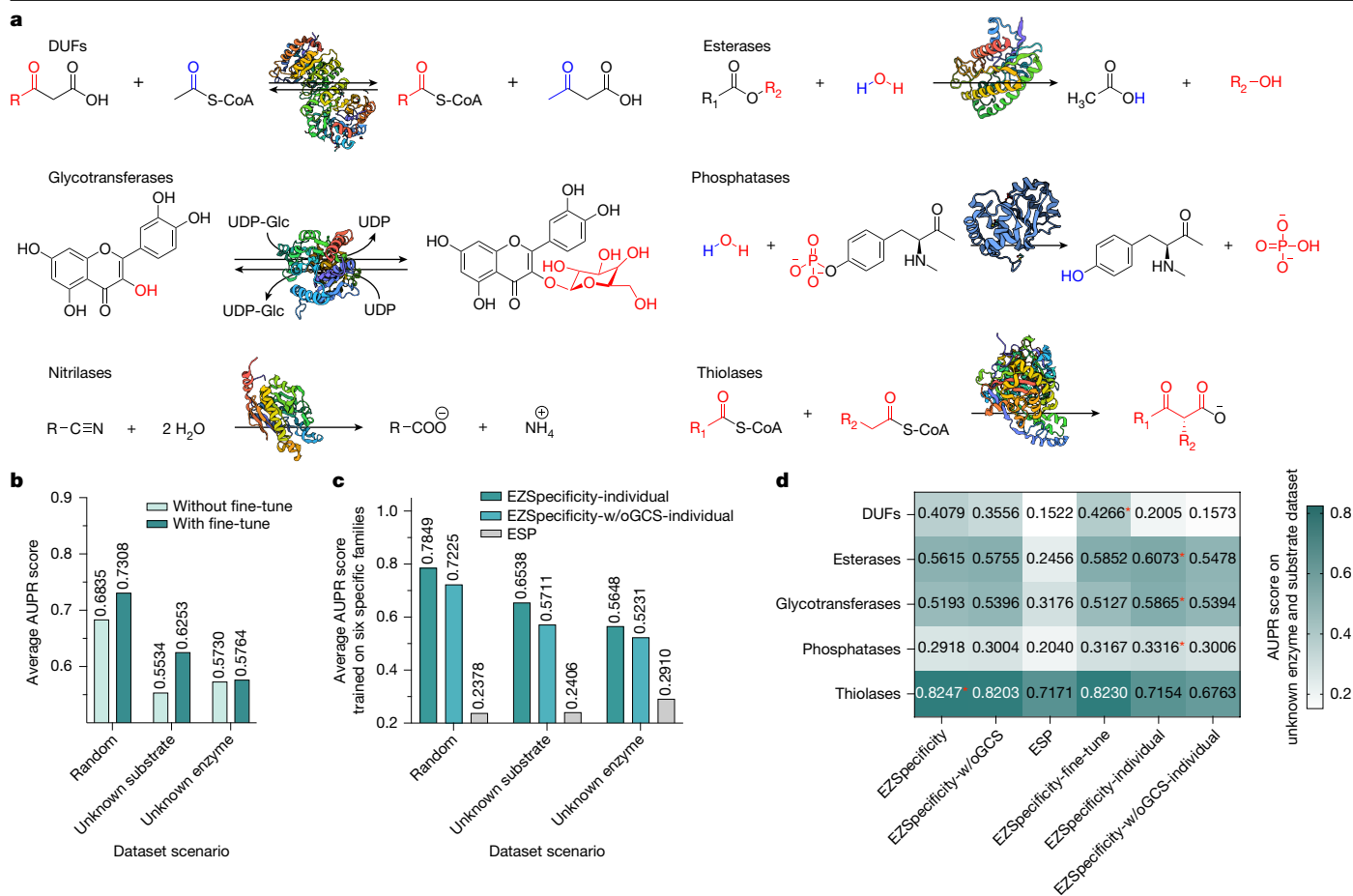


Fig. 4 | In silico evaluation of EZSpecificity using six representative enzyme families. **a**, The enzymatic functions of six enzyme families. The structure of a representative enzyme from each family is shown. **b**, The averaged AUPR performance of EZSpecificity with and without fine-tuning on six enzyme families. **c**, The average AUPR performance of the EZSpecificity-individual model across six enzyme families for three types of dataset splits. **d**, The AUPR

performance of EZSpecificity on specific protein families under the unknown enzyme and substrate data split setting. Nitrilases were excluded from this evaluation because of the extremely low number of data points, which rendered the results highly unreliable. For instance, no positive pairs may have occurred during the four-fold validation.

α -ketoglutarate-dependent halogenases, flavin-dependent halogenases and SAM-dependent halogenases; Fig. 5a–c and Supplementary Figs. 16–21). Notably, EZSpecificity without any fine-tuning achieved 0.7720–0.9447 AUROC and 0.5430–0.8506 AUPR values, varying from easy to hard dataset scenarios (Fig. 5d–f). As observed in six proof-of-concept enzyme families (Fig. 4b), the fine-tuning process further enhanced the prediction performance (Fig. 5d–f), and EZSpecificity-fine-tune achieved 0.8008–0.9600 AUROC and 0.5698–0.8823 AUPR values. Specifically, the AUPR of EZSpecificity-individual increased from 27.87% to 35.26% compared with EZSpecificity-w/oGCS-individual, as the task difficulty increased from random to unknown enzyme and substrate scenarios. These results strongly suggest that the integration of structural features, graph and cross-attention mechanisms contributes to the improved performance of EZSpecificity.

Among all halogenases, eight flavin-dependent halogenases, known for their distinct regioselectivity within the four major halogenase families, were selected for experimental validation along with 78 substrates. The average molecular similarity between the 78 substrates and 449 collected substrates from the literature is only about 9% (Supplementary Fig. 18), highlighting the significant diversity among these substrates. These enzymes were selected based on the following criteria: (1) they have been reported to exhibit broad substrate scope; (2) they can be well-expressed as soluble proteins in *Escherichia coli*; (3) they are carrier protein-independent halogenases; and (4) they have either experimentally determined structures or reliable predicted

structures from AlphaFold (for example, Th-Hal and AetA; Supplementary Fig. 22). Using a high-throughput screening platform configured with 96-well plates and a liquid chromatography–mass spectrometry detection method, a total of 624 experimental halogenation data points were collected. Then, EZSpecificity, trained on the in-house halogenase HaloS dataset (about 3,300 data points in literature extracted by the semi-automatic data extraction approach; for details, see Supplementary Text 1), was directly applied to predict patterns of halogenase–substrate interactions. EZSpecificity, EZSpecificity-w/oGCS and ESP each generated a list ranking eight enzymes for each substrate. The accuracy of these recommendations would serve as the basis for comparison. As confirmed by experimental results, 55 of them did not have any enzymes that could act on them (Supplementary Figs. 22–26, 28–48 and Supplementary Table 8). For the 12 new substrates not previously seen in the database (Supplementary Fig. 49), the EZSpecificity-fine-tune model, which was fine-tuned using the HaloS dataset, achieved a prediction accuracy of 91.7% for its top 1 recommendation (Fig. 5e). This significantly surpassed the baseline methods, including EZSpecificity-w/oGCS (41.7%) and ESP (58.3%), by a considerable margin. Interestingly, in our practical tests (Fig. 5e), the accuracy of the EZSpecificity-individual model (66.7%) was lower than that of the EZSpecificity-fine-tune model. Moreover, we introduced the EZSpecificity-ensemble model, which is an ensemble of the EZSpecificity-individual and EZSpecificity-fine-tune models, achieving an accuracy of 75.0%. Notably, the prediction accuracy of various

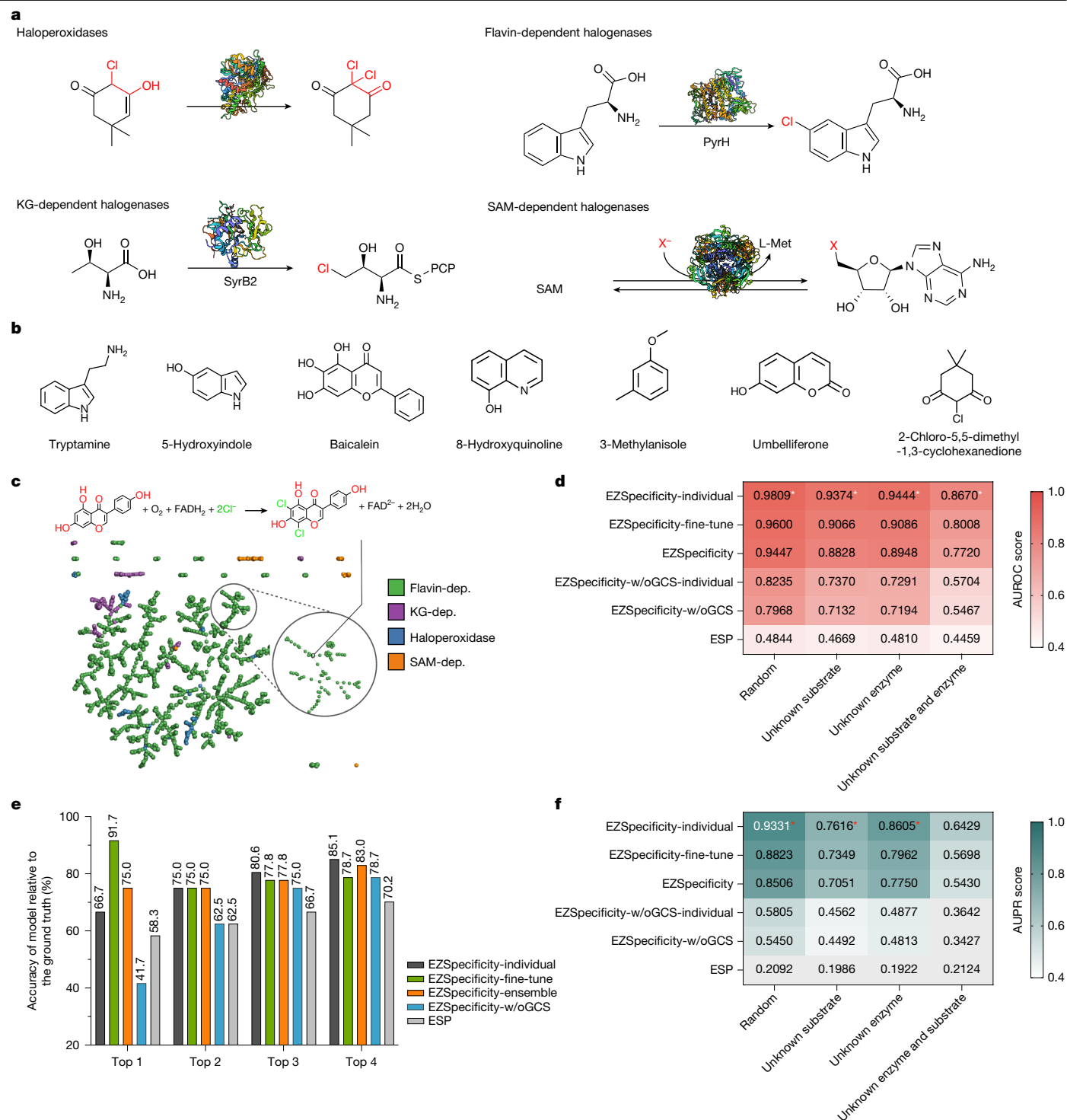


Fig. 5 | In silico and in vitro experimental validation of EZSpecificity on the in-house HaloS dataset collected by a semi-automatic data extraction approach. a, The halogenation function of four types of halogenases.

The typical structure of each halogenase is shown. X stands for halogen atom (Cl⁻, I⁻ and F⁻). **b**, The structure of representative collected substrates in HaloS dataset. **c**, TMAPs of halogenation reaction in HaloS dataset, colour-coded on four types of halogenases. TMAP is a data visualization method, which is capable of representing datasets of up to millions of data points and

arbitrary high dimensionality as a two-dimensional tree (<http://tmap.gdb.tools>). **d,f**, Comparison between EZSpecificity and state-of-the-art machine learning models for predicting enzyme substrate specificity on halogenases for AUROC (**d**) and AUPR (**f**) scores. The highest value was labelled with an asterisk in white or red colour. **e**, The prediction accuracy of EZSpecificity and the state-of-the-art machine learning models for predicting eight enzyme substrate specificity relative to the ground truth.

EZSpecificity models relative to ground truth was maintained when the scope was broadened to include the top 2–5 recommended enzymes, in which it ranged between 62.5% and 85.2% (Fig. 5e, Supplementary Fig. 27 and Supplementary Table 8).

Representative applications

Because of its ability to predict a score for a target enzyme–substrate pair, EZSpecificity can be used to (1) rank all the potential substrates

for a target enzyme based on substrate specificity; and (2) rank all the enzyme(s) for a target substrate based on substrate specificity. As a case study, 34 metabolites and 860 enzymes present in *E. coli* were selected for investigation. After generating 29,240 = 34 × 860 docking structures, we challenged EZSpecificity to identify the enzyme(s) that act on each target metabolite. As shown in Extended Data Fig. 1a and Supplementary Text 4, among the 34 metabolites, EZSpecificity successfully matched 10 metabolites (29.4%) with their corresponding enzymes within the top 5% of ranked enzymes compared with using docking scores, which achieved 20.4%. When expanding the ranking threshold to the top 20%, the cumulative success rate increased to 50% (17 metabolites). As shown in Extended Data Fig. 1b, a two-dimensional kernel density estimation plot was generated to examine the relationship between the percentile rank of the top predicted enzymes and their sequence similarity to enzymes in the training set. Analysis of the percentile rank of top enzyme prediction showed a peak density at approximately 10%, indicating that EZSpecificity demonstrates the highest confidence in selecting the correct reactive enzyme when it is ranked in the top 10% of candidates. Also, a notable density peak is observed in the bottom-left region, suggesting that EZSpecificity performs well on similar sequences while the model maintains reasonable predictive accuracy even for enzymes with low sequence similarity to those encountered during training, demonstrating its capacity for generalization to new sequences.

Furthermore, EZSpecificity holds the potential for the study of biosynthetic gene clusters (BGCs) by linking BGC genes to their corresponding biosynthetic intermediates. This ability was validated using two representative BGCs: the clavulanic acid biosynthetic pathway^{41–43} and the albonoursin biosynthetic pathway^{44,45} (Extended Data Fig. 2). EZSpecificity achieved up to 66.7% accuracy in identifying the correct target enzyme among the top three ranked candidates for each step in the biosynthetic pathway (Supplementary Tables 10, 11, 14 and 15 and Supplementary Text 5). Identifying gene-intermediate pairs within BGCs remains the main challenge⁴⁶, as intermediates often differ by subtle chemical modifications. The performance of EZSpecificity could be further improved through training on additional BGC-relevant data, although these datasets remain poorly organized at present. Moreover, Supplementary Tables 12 and 13, Supplementary Fig. 51 and Supplementary Text 6 further proved the practical utility of EZSpecificity by examining the specific enzyme families.

Taken together, these case studies demonstrate the utility of EZSpecificity for highly efficient functional characterization of metabolomes and natural product libraries. Notably, EZSpecificity focuses on substrate specificity prediction. Although enzyme stereoselectivity can be processed during the encoding process, EZSpecificity may not be able to distinguish its subtle differences. This limitation arises because our current GNN encoding method treats different stereoselectivity at the same atom equally.

Conclusions

In conclusion, we developed a general deep learning model, EZSpecificity, for predicting enzyme substrate specificity, which can be used across diverse protein families and native and non-native substrates. This model introduces two innovations—an encoder for 3D binding complexes based on the ESIBank dataset and cross-attention layers, both of which are often overlooked in previous CPI studies—addressing the limitations of existing approaches that often overlook the 3D binding process and atomic enzyme–substrate interactions. The ESIBank complexes also provide a valuable community resource for studies of enzyme function and reactivity, although performance may be affected for pairs lacking active site annotations. Apart from this, EZSpecificity demonstrated strong performance, up to 48.1% higher AUROC than ESP in silico 91.7% accuracy compared with 58.3% for ESP in vitro across eight halogenases and 78 substrates.

EZSpecificity thus represents a broadly applicable tool for biocatalysis^{47,48}, synthetic biology⁴⁹, enzyme engineering^{50,51} and drug discovery^{52,53}, with future integration of dynamic binding information poised to further enhance predictive power.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09697-2>.

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Methods

ESIBank database preparation

To build the ESIBank, we first collected existing native and/or non-native substrates from BRENDA⁵⁴ in SMILES. However, in BRENDA, the specific enzyme sequences corresponding to substrates are mostly not directly listed. Instead, only the EC number and the organism associated with the enzymes are provided. Therefore, we randomly selected an enzyme from the listed category, comprising EC number and organism name, to match the substrate obtained from UniProt²⁹, resulting in approximately 180,000 positive enzyme–substrate pairs. To generate negative samples, for each positive pair, we created five negative enzymes and five negative substrates, totalling 10 negative enzyme–substrate pairs per positive pair. These negative pairs vary in difference with the actual enzyme and substrate to help the model cover all resolutions of enzyme–substrate specificity prediction. The digit number in EC numbers was used to define the levels of difference between positive and negative data. These levels ranged from no common digits in the EC number to all four digits being identical, encompassing five levels. As a result, using EC numbers as a guide, we randomly generated approximately 1,100,000 negative substrate–enzyme pairs across these five different levels, corresponding to the different digits of the EC number. Some abnormal or low-quality data points were removed, including substrates containing more than 280 atoms, enzymes containing more than 1,000 amino acids and enzymes lacking information about active sites in UniProt. Moreover, additional data from six representative enzyme families, including esterases²⁴, glycotransferases²⁵, nitrilases²⁶, phosphatases²⁷, thiolases⁶ and domain of unknown function proteins (DUFs)²⁸, were included to enrich ESIBank. Moreover, about 3,300 enzyme–substrate pairs from halogenases were extracted using a semi-automatic data extraction process consisting of four steps: identification, extraction, translation and connection. The detailed process is provided in Supplementary Text 1. In total, we collected 323,783 high-quality enzyme–substrate pairs (Supplementary Tables 1 and 7).

Construction of 3D enzyme–substrate complexes

To make the ESIBank more informative, we generated the prediction of 3D enzyme–substrate complexes of each enzyme–substrate pair. Previous methods often used a brute force approach to dock substrates directly to enzymes, resulting in inaccuracies due to the inability of the current docking software to precisely identify the enzyme-binding sites⁵⁵. Instead, we sourced active site information from the UniProt database and constrained the binding box to a 20 Å cube centred around the active site amino acids because of the size of the dataset. Moreover, inaccuracies stemming from the absence of cofactors in enzyme structures were addressed in our pipeline by integrating the cofactors into the enzyme structures using AlphaFill⁵⁶. This ensures that the catalytic enzyme–substrate pose can be generated in a local environment similar to that of the native binding pocket.

Specifically, substrate structures were generated from SMILES strings using Open Babel⁵⁷ and RDKit⁵⁸. The 3D structures of reactive enzymes, along with any required cofactors, were obtained from the AlphaFill database⁵⁶. All apo enzyme models were sourced from the FTP archive of the AlphaFold Protein Structure Database⁵⁹, following the procedures outlined in the AlphaFill documentation. After the preparation of all input enzyme and substrate files, docking was performed on all enzyme–substrate pairs in the ESIBank using AutoDock-GPU^{31,60}. First, all enzyme structure files were converted to map files for input into AutoDock-GPU and the substrate files were converted to pdbqt inputs. Next, a list of all enzymes in the dataset with known active sites was generated. For all enzymes with known active sites, the centre of mass of the active site was used as the centre of the box. For all enzymes without a known active site, the centre of the enzyme was used as the box centre. To leverage available active site information, a box cutoff

radius of 20 Å from the centre of the active site was applied for enzymes with known active site coordinates. During docking, the number of runs was set to 100 per enzyme–substrate pair, in which each run consisted of a maximum of 2,500,000 score evaluations for a unique binding position for a total maximum of 250 million pose evaluations. After docking, the highest scoring pose was produced as output as a structure file, and final pose predictions were used to generate structure files of the enzymes with substrate bound for input into the model. Of course, other efficient docking software or complex generation tools, such as Co-Dock⁶¹ and AF3⁶² could also be explored further.

Use of a pretrained protein language model to represent enzymes

We used ESM-2 (refs. 8,13,63), a pretrained transformer protein language model, to embed the protein amino acid sequences in ESIBank. ESM-2 is one of the most powerful self-supervised protein language models, having 15 billion parameters, which was trained to reconstruct the portions of proteins that are randomly masked during training over protein sequences in the Uniref database. We used the output from the penultimate layer of the neural network as the amino acid embedding for enzymes. The embedding generated by ESM-2 is a 1,280-dimensional vector representation for each amino acid. A linear layer with 128-dimensional embedding was used for each amino acid sequence, which was reprojected from the original 1,280-dimensional vector in ESM-2. We referred to it as the amino acid representation of the enzyme and the average of these as the representation of the enzyme.

Capturing catalytic pocket environment by SE(3)-equivariant GNNs

Each atom of the substrate was encoded into a vector representation to represent its surrounding micro-environment using an SE(3)-equivariant GNN²³. We represented the binding pocket as graphs $\mathbf{G} = (\mathbf{V}, \mathbf{E})$, where each node $\mathbf{V}$ is an atom from either substrate or enzyme. These nodes are connected by edges in $\mathbf{E}$ to their k -nearest atom neighbour in the structure. In practice, we set k as 32. The features of enzyme atoms include chemical elements, amino acid types and an indication of whether the atoms belong to the backbone. In the case of substrate atoms, their types are represented as multi-hot vectors, which include information about chemical element types and aromatic properties. The edge features e_{ij} include distance embeddings and bond types. For distance embeddings, we use radial basis functions positioned at 32 centres within the range of 0–10 Å. Bond types are represented by a four-dimensional one-hot vector, indicating whether they are single, double, triple or virtual bonds, along with a two-dimensional one-hot vector to specify whether they connect atoms between substrates and enzymes.

To make the encoding process equivariant with respect to E(3) transformations, inspired by E(n)-equivariant graph neural networks²³, we modelled the catalytic site environment with an SE(3)-equivariant GNN:

$$h^t = \varphi_{\theta}(h^0, e_{ij})$$

At the l th layer, the atom hidden embedding h^l is updated as follows:

$$h_i^{l+1} = h_i^l + \varphi_h \left(\sum_{j \in \mathcal{N}(i)} \varphi_e(e_{ij}, h_i^l, h_j^l) \right),$$

where φ_h and φ_e are two-layer perceptrons to model message aggregation and message passing separately. The initial hidden embedding h^0 is obtained by two distinct linear layers, each used to reproject the features of enzyme atoms and substrate atoms independently. We referred to the final atom hidden embedding h^t as the micro-environment embedding, the final atom hidden embedding for substrate atoms as substrate atom embedding, and the average of substrate atom embedding as the substrate embedding.

Modelling the interactions between substrates and enzymes using cross-attention layers

Existing ML models usually use a multi-layer perceptron⁶⁴ over the concatenation of substrate embedding and enzyme embedding to model the enzyme–substrate interactions⁷⁸. However, this design cannot directly model the atomic interaction, which plays an important part in determining enzyme specificity. To overcome this limitation, inspired by Transformer⁶⁵, we used a cross-attention layer to generate enzyme-aware substrate atom embedding:

$$\text{Attention}(Q, K, V) = \text{soft max} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

$$\text{Cross-attention}(E, S) = \varphi_o(\text{Concat}(\text{head}_1, \dots, \text{head}_h))$$

where $\text{head}_i = \text{Attention}(\varphi_{q_i}(E), \varphi_{k_i}(S), \varphi_{v_i}(S))$, E is the enzyme amino acid embedding, S is the substrate atom embedding, d_k is the dimension of the embedding K and φ is the linear layer. As shown in the equations, the cross-attention layer contained h independent heads. Each of these heads is a learnable attention function to capture the interaction between the amino acids of enzymes and the atoms of substrates. This function is composed of several trainable linear layers. It first generates an attention matrix according to the embeddings of amino acids and atoms, and then performs a weighted summation of atom embeddings. Similarly, we used another cross-attention layer to generate substrate-aware enzyme amino acids embedding:

$$\text{Cross-attention}(S, E) = \varphi_o(\text{Concat}(\text{head}_1, \dots, \text{head}_h))$$

where $\text{head}_i = \text{Attention}(\varphi_{q_i}(S), \varphi_{k_i}(E), \varphi_{v_i}(E))$.

We referred to the average embeddings of enzyme-aware substrate atom embedding as enzyme-aware substrate embedding and the average embeddings of substrate-aware enzyme amino acids embedding as substrate-aware enzyme embedding.

Multilayer perceptron as the base predictor

Enzyme and substrate are separately encoded by ESM-2 and SE(3)-equivariant GNN²³. We denote enzyme embedding as h_{enzyme} and substrate embedding as $h_{\text{substrate}}$. Through cross-attention layers, we directly modelled the interactions between the amino acids of the enzyme and the atoms of the substrate, enriching their embeddings with each other's context. This process resulted in the creation of substrate-aware enzyme embedding $\bar{h}_{\text{enzyme}}$ and enzyme-aware substrate embedding $\bar{h}_{\text{substrate}}$. These features were used as input to a multi-layer perception to predict the enzyme specificity:

$$y = \text{MLP}(\text{Concat}(\bar{h}_{\text{substrate}}, \bar{h}_{\text{enzyme}}, h_{\text{substrate}}, h_{\text{enzyme}}))$$

Training process of EZSpecificity

We formulated the task of predicting enzyme specificity as a binary classification problem, with the training objective being the classification of pairs of substrates and enzymes into two categories. Our model was trained using a cross-entropy loss function, and we employed the AdamW optimizer with default parameters⁶⁶. The learning rate was set at 0.0003. To enhance training stability, we implemented a learning rate schedule. In the initial few epochs, we linearly increased the learning rate from 0.000006 to 0.0003 for warm-up purposes. If the model's performance did not improve for 10 consecutive epochs, we reduced the learning rate to half of its previous value. Training was concluded once the learning rate dropped below 0.000006. We set the batch size to 32 for training. The SE(3) neural network comprised three GNN layers and the base predictor, multi-layer perceptron, consisted of three

feedforward layers. Unless specified otherwise, the dimensions of the hidden embeddings were all set to 128.

Chemicals and strains

All chemicals were of analytical grade or higher quality and purchased from Sigma-Aldrich and Fisher Scientific and used as received unless stated otherwise. Genes encoding various halogenases, including PrnA (UniProt ID: P95480), PyrH (UniProt ID: A4D0H5), AetA (UniProt ID: A0A861B8S3), BorH (UniProt ID: M9QSI0), ThaL (UniProt ID: A1E280), BrvH (UniProt ID: B4WBL8), PltM (UniProt ID: Q4KCZ3) and Th-HaL (UniProt ID: Q4KCZ3), were synthesized by Twist Bioscience and cloned into the pET28a(+)-vector by *NcoI* and *XhoI* restriction sites with a C-terminal His₆-tag. Chemically competent *E. coli* BL21-Gold (DE3) cells (New England BioLabs) were used as hosts for protein expression.

Expression and purification of experimentally validated halogenases

All experimentally validated halogenases (PrnA, PyrH, AetA, BorH, ThaL, BrvH and PltM) were expressed as described elsewhere¹³. After enzyme expression induced by adding 0.5 mM isopropyl β -D-1-thiogalactopyranoside (IPTG) and continuing the incubation at 30 °C for 18 h, the cells were collected by centrifugation at 4 °C and $3,600 \times g$ for 15 min. The resulting cell pellets were resuspended in 20 ml of buffer A, which contained 50 mM sodium phosphate (pH 8.0) and 10 mM imidazole. Sonication was performed for 10 min, alternating between 10 s on and 5 s off at 50% amplitude, to disrupt the cells. Cellular debris was then removed by centrifugation at $14,000 \times g$ for 1 h. Protein purification was carried out using a Protino Ni-IDA 2000 packed column as per the instructions of the manufacturer using buffer A and buffer B (50 mM sodium phosphate, 330 mM imidazole, pH 8.0). To remove imidazole, a PD-10 desalting column (GE Healthcare) was used. The purified enzymes were stored in small aliquots in sodium phosphate buffer (50 mM sodium phosphate, pH 8.0, 5% (v/v) glycerol) at -80 °C, ensuring that each aliquot was used only once post-thawing. The purity of the proteins was confirmed by SDS-PAGE (sodium dodecyl sulfate–polyacrylamide gel electrophoresis), showing greater than 90% purity, and the protein concentration was quantified using the Pierce BCA protein assay. To rule out the possibility of non-regiospecific or non-enzymatic halogenation caused by H₂O₂ or HOBr leakage, we supplemented the reaction with superoxide dismutase (3 U) and catalase (3 U) as controls for comparison. These enzymes specifically eliminate reactive oxygen species, ensuring that any halogenation observed is due to the enzymatic activity, rather than uncontrolled chemical reactions driven by leaked oxidizing agents. As shown in Supplementary Figs. 29, 33, 35, 37 and 38, identical halogenation results were observed in both the presence and absence of superoxide dismutase and catalase, confirming that the halogenation was catalysed enzymatically, rather than being the result of non-enzymatic reactions.

High-throughput enzymatic assays and product identification with high-resolution mass spectrometry

The enzymatic reactions were conducted in a 96-well plate with shaking at 900 rpm. Reactions (100 μ l) consisting of 1 mM target substrate, 10 μ M FAD, 1 mM NADP⁺, 10 mM sodium phosphate, 10 μ M purified enzymes (PrnA, PyrH, AetA, BorH, ThaL, BrvH, Th-HaL and PltM), 5 μ M SsuE and PtdH were incubated for 24 h and 30 °C with and without NaBr in a buffer consisting of 50 mM sodium phosphate (pH 8.0). Aliquots (100 μ l) were taken out and mixed with 50 μ l ice-cold 2% formic acid to terminate the reaction by precipitating the proteins. Precipitates were removed by centrifugation ($15,000 \times g$ for 5 min at 4 °C) before 5 μ l aliquots were subjected to liquid chromatography–mass spectrometry (LC-MS). All LC-MS analyses were carried out on a Thermo Scientific LC-MS, using a Hypersil GOLD VANQUISH PFP UHPLC column (1.9 μ m, 2.1 mm $\times$ 100 mm) at a flow rate of 0.4 ml min^{−1} with the following elution system: solvent A (H₂O supplemented with 0.1% formic acid) and

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B (acetonitrile supplemented with 0.1% formic acid) and a specific linear gradient for various substrates. ThermoFisher Scientific Q Exactive was equipped with an ESI source and Orbitrap mass analyser. Calibration was performed with Pierce LTQ Velos ESI Positive Ion Calibration Solution (ThermoFisher). Xcalibur from ThermoFisher Scientific was used to control, acquire and interrogate data from the LC-MS system and related instruments. The full MS-Selected Ion Monitoring (SIM) was operated with the following parameters: 70,000 (resolution), 50–750 *m/z* (scan range), $T = 3 \times 10^6$ (AGC target), 200 ms (maximum injection time) and positive (polarity). Data analysis was then conducted using the Freestyle application within Xcalibur software. Product identification was based on retention time comparison and the detection of an increased molecular weight marked by bromine.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The ESIBank dataset can be found at https://drive.google.com/drive/folders/1-lfpKPhmf_P837UBfgUt_7R00_f45DtA?usp=drive_link.

Code availability

The source code of EZSpecificity is available at Zenodo⁶⁷.

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Author contributions H.C., Y.S. and H.Z. conceived of the presented idea. H.C., Y.S., T.Y. and T.J.D. designed the ESIBank database collection process. H.C. planned the in vitro experiments, and H.C. and Z.Z. carried out in vitro experiments and data analysis. Y.S. developed and implemented the computational framework. H.C. and Y.S. designed the in silico experiments and data analysis. H.C. and T.J.D. contributed to the docking process. H.C. contributed to specific families and halogenase data collection and preparation. T.Y. contributed to the 3D structure prediction and code revision. H.C. wrote the paper with inputs from all authors. All authors discussed the results and contributed to the final paper. H.Z., D.S. and J.P. provided supervision and resources for this study.

Competing interests The authors declare no competing interests.

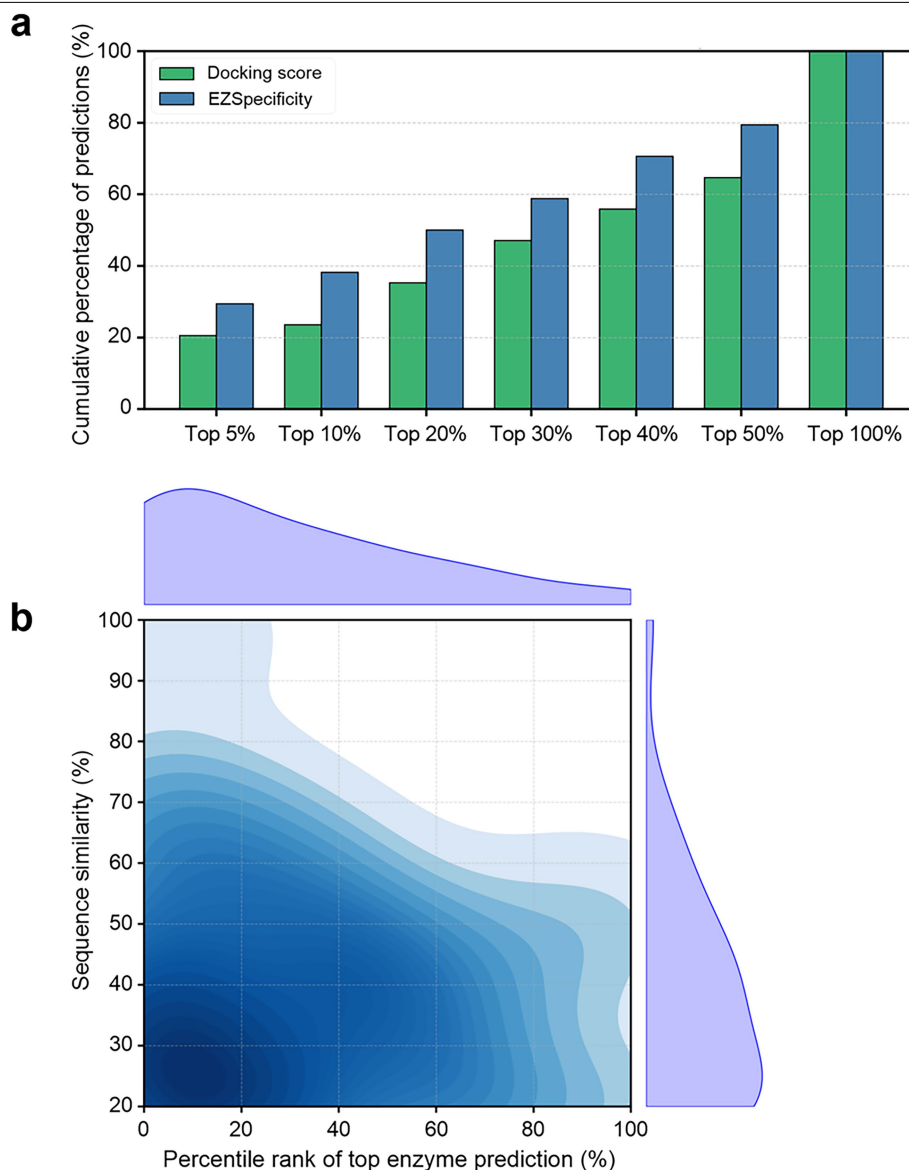
Additional information

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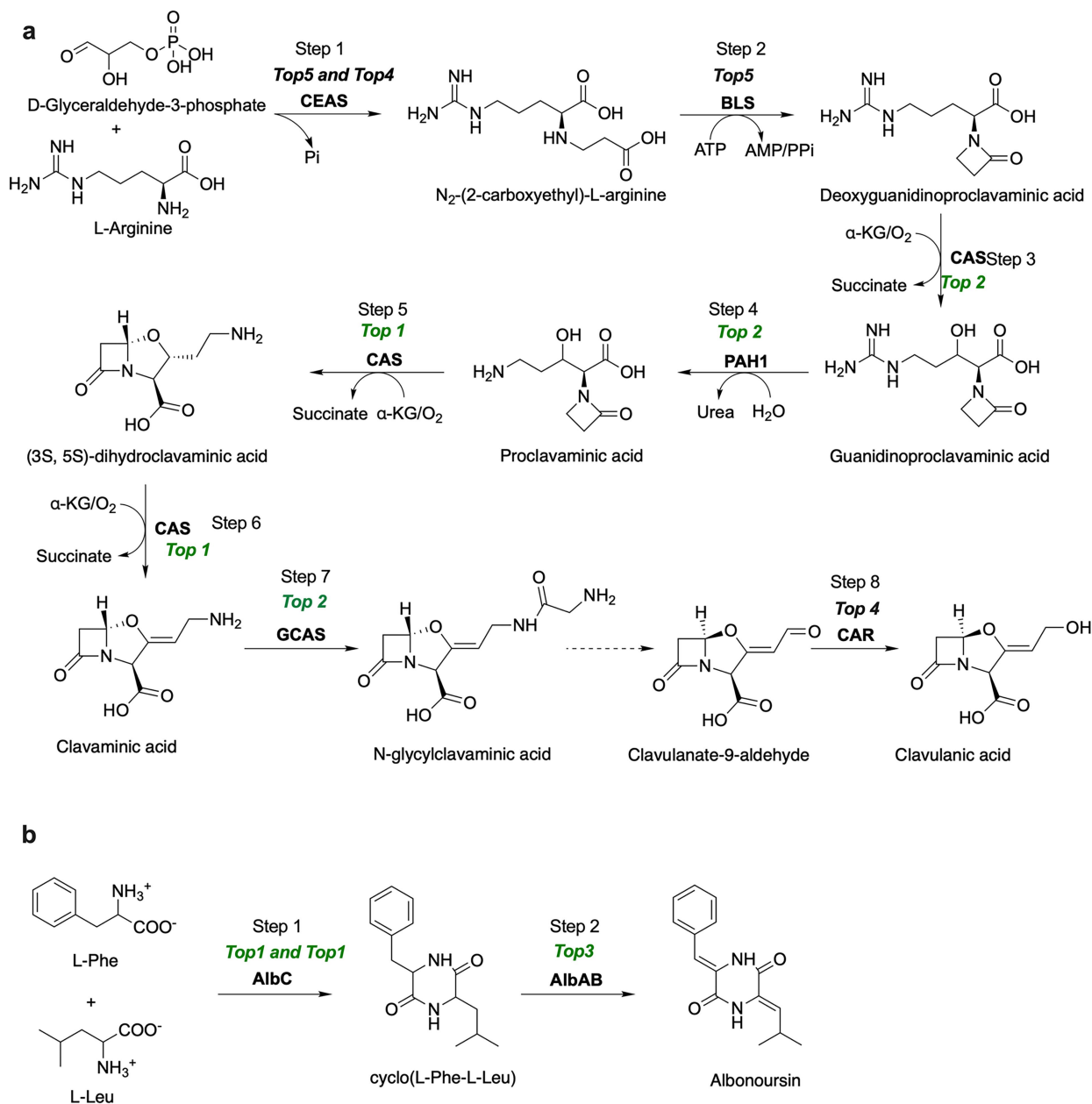
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Extended Data Fig. 1 | EZSpecificity examination in the metabolite-enzyme pair prediction task. (a) The performance of EZSpecificity *vs.* Docking Score in identifying the reactive enzyme(s) for 34 target metabolites. AutoDock Vina was used as a representative physical model for docking scores. Only one reactive enzyme with the best-ranked score was considered for calculation. EZSpecificity predicts 29% of the known positive enzyme within the top 5% of predictions compared to 20% from physical score alone. (b) 2D kernel density estimation plot of the percentile rank of top predicted enzymes *vs.* the sequence

similarity compared to any enzyme in our training set for EZSpecificity in the metabolite-enzyme pair prediction task. Kernel density estimation analysis can answer a fundamental data smoothing problem where inferences about the population are made based on a finite data sample. 1D kernel density estimation for each axis was also plotted at the top and right of the figure. The data was obtained from 34 metabolites within *E. coli* used for the calculation. Density is indicated by color intensity, where darker regions correspond to greater density.



Extended Data Fig. 2 | Enzyme retrieval by EZSpecificity within two representative BGCs. (a) Diagram of the clavulanic acid BGC, comprising eight reaction steps. **(b)** Diagram of the albonoursin BGC, comprising two reaction steps. Two substrates were investigated in the first reaction step in each BGC.

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